

Figure 1: Virtual Tenant Network

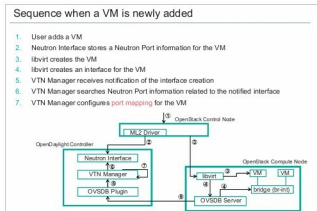


Figure 2: Request flow

To recap, I said that with the help of the VTN Manager, the user can group multiple ports from multiple switches in the infrastructure to form a single isolated network.

Let's compare this with our OpenStack environment, which has two Open vSwitches installed in the controller and compute node.

1. Whenever a new network is created in OpenStack, VTN Manager creates a new VTN in ODL.
2. Whenever a new sub-network is created, VTN Manager handles it and creates a vBridge under the VTN. vBridge is nothing but the virtual switch.
3. When a new VM is created in OpenStack, the addition of a new port in the Open vSwitch of the compute node is captured by VTN Manager, and it creates a vBridge

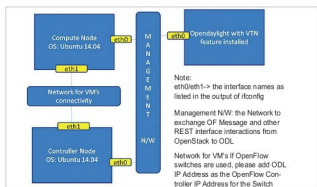


Figure 3: LAB layout

interface in the newly created vBridge and maps that Open vSwitch port with the particular vBridge port.

4. In this case, the port (say, vs1p1) of the DHCP agent in the Open vSwitch of the controller node and the port (vs2p1) of the VM in the compute node are isolated from the actual Open vSwitch, using the flow entries from the OpenDaylight VTN Manager, to form a new virtual switch environment called the virtual tenant network.
5. When the packet sent from the DHCP agent reaches the OpenStack controller's Open vSwitch port vs1p1, then flow entries will tell the port vs1p1 to forward the packet to the compute node's Open vSwitch port vs2p1 using the underlying physical network. This packet will be sent as a regular TCP packet with a source and destination MAC address, which means that the traffic created in one network can be sent as a regular packet across the controller and compute node without any tunnelling protocol.
6. This explains why support for VxLAN and GRE network types is not required.

LAB set-up layout

The VTN features support multiple OpenStack nodes. Hence, you can deploy multiple OpenStack compute nodes.

In the management plane, OpenDaylight controller, OpenStack nodes and OpenFlow switches (optional) should communicate with each other.

In the data plane, Open vSwitches running in OpenStack nodes should communicate with each other through physical or logical OpenFlow switches (optional). Core OpenFlow switches are not mandatory. Therefore, you can directly connect to the Open vSwitches.

You may need to disable the firewall (UFW) in all the nodes to reduce the complexity.

Installing OpenStack with the Open vSwitch configuration

Installing OpenStack is beyond the scope of this article; however, getting started with a minimal multi-node OpenStack deployment is recommended.

To help speed up the process, you could use my fully automated bash script for installing the OpenStack-Mitaka set-up at <https://github.com/CloudenablersPvtLtd/openstack-setup>.

Note: This script will install OpenStack and configure the Linux bridge for networking. But for the VTN integration to work in OpenStack, we need network configuration with Open vSwitch. So, you must uninstall the Linux bridge settings and reconfigure with Open vSwitch.

After the successful OpenStack installation, run the sanity test by performing the following operations.

Create two instances on a private subnet. Then add the floating IP address from your public network, verify that you can connect to them and that they can ping each other.